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/*
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PracticalNo:3
*/
#include<iostream>
using namespace std;
#define MAX 5
int stack[MAX];
int top=-1;
void push(){
int order;
if(top == MAX-1){
 cout<<"Stack Is Full \n";
}
else{
 cout<<"Enter Next Order:";
 cin>>order;
 top++;
 stack[top]=order;
}
}
void display(){
 if(top == -1){
 cout<<"Stack is Empty \n";
 }
 else{
 cout<<"Displaying Stack:";
 for(int i=top;i>=0;i--){
 cout<<stack[i]<<" ";
 }
 }
}
void pop(){
if(top == -1){
 cout<<"\nStack is Empty \n";
}
else{
 cout<<"\norder "<<top<<" is Cancelled\n";
 top--;
}
}
void peek(){
 if(top==-1){
 cout<<"Stack Is Empty";
 }
 else{
```

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 cout<<"Recent Order :"<<stack[top];
 }
}
void isempty(){
 if(top == -1){
 cout<<"Stack Is Empty";
 }
 else{
 cout<<"Stack is Full";
 }
 }

void isfull(){
 if(top == MAX-1){
 cout<<"Stack overflow";
 }
 else{
 cout<<"Stack underflow";
 }
 }

int main(){
int choice;
do{
 cout<<"\n";
 cout<<"\n1.Add Your order";
 cout<<"\n2.Canceling Last order";
 cout<<"\n3.Display last order";
 cout<<"\n4.Orders are full";
 cout<<"\n5.Orders are empty";
 cout<<"\n6.Display All orders";
 cout<<"\n7.Exit";

 cout<<"\nEnter Your Choice:";
 cin>>choice;

 switch(choice){
 case 1:
 push();
 display();
break;

 case 2:
 pop();
 display();
 break;

 case 3:
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 peek();
 display();
 break;

 case 4:
 isfull();
 display();
 break;

 case 5:
 isempty();
 display();
 break;

 case 6:
 display();
 break;

 case 7:
 cout<<"Welcome";
 break;

 default:
 cout<<"Exitting";
 }
 }
 while(choice!=7);

return (0);
}
/*
(base) pl-12@pl12-OptiPlex-3020:~$ ^C
(base) pl-12@pl12-OptiPlex-3020:~$ g++ 183.cpp
(base) pl-12@pl12-OptiPlex-3020:~$ ./a.out
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
6.Display All orders
7.Exit
Enter Your Choice:3
Stack Is EmptyStack is Empty
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
```

```
6.Display All orders
7.Exit
Enter Your Choice:645
Exitting
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
6.Display All orders
7.Exit
Enter Your Choice:23
Exitting
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
6.Display All orders
7.Exit
Enter Your Choice:1
Enter Next Order:5
Displaying Stack:5
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
6.Display All orders
7.Exit
Enter Your Choice:1
Enter Next Order:2
Displaying Stack:2 5
1.Add Your order
2.Canceling Last order
3.Display last order
4.Orders are full
5.Orders are empty
6.Display All orders
7.Exit
Enter Your Choice:^C
(base) pl-12@pl12-OptiPlex-3020:~$
*/
```